

CHANGING SEA LEVELS

Sea levels are always rising or falling in relation to land. Ice ages affect sea levels by locking up water on continents as ice, and then releasing it into the oceans when the ice melts. The land rises when the heavy ice melts, and over longer periods it can be pushed upward by moving plates in the Earth's crust. As a result, sedimentary rocks formed on the sea floor are now found on land, and many coasts have features showing that the land has been submerged.



◀ RAISING THE ROCKS

Where sedimentary rocks have been uplifted, erosion often reveals the layers that were laid down on the ocean floor, along with the fossils preserved within them. Massive ground movements often buckle and fold these layers, as on this rocky coast.



CRETACEOUS OOZE

The pure white limestone known as chalk is made of the skeletons of millions of microscopic marine organisms. Known as coccoliths, they settled on the floors of tropical oceans during the Cretaceous Period, about 100 million years ago, forming layers of biogenic ooze. The soft ooze was compressed into chalk up to 1,300 ft (400 m) thick, then raised by ground movements to form the rolling hills and white cliffs of regions such as southern England, shown below.



MICROSCOPIC
COCCOLITH



► DROWNED LANDSCAPES

After the last ice age, sea levels rose to roughly their current point, drowning ancient landscapes. Deep, U-shaped glacier valleys were flooded to create steep-sided fjords like this one in Scandinavia.

